

TEAM ID	SWTID-2026-7349
PROJECT TITLE	India's Agricultural Crop Production Analysis using Tableau
MAX MARKS	5 MARKS

4.2 – Proposed Solution

Indias agricultural crop production analysis using Tableau is a deal. It is supposed to help us understand the sector, which is very important for the supply chain. This sector is going to change a lot in the coming years. The main goal is to go beyond just looking at data and create a system that can predict crop production and find factors that affect it.

To do this we use tools. We start by cleaning the data using Microsoft Excel. We remove any values. Then we use Python to find any missing values or duplicates. The hard work is done in Tableau Desktop, where we create dashboards to show production trends, seasonal impacts and how different regions are doing.

These dashboards are the center of our solution. They are designed to give us useful insights. We can see how production has changed over the years compare production and find out which states are doing well.

For example, Kerala is doing great because of its coconut production. We also look at how area and production efficiencies related. We found out that crops like coconut can produce a lot even when they do not use area.

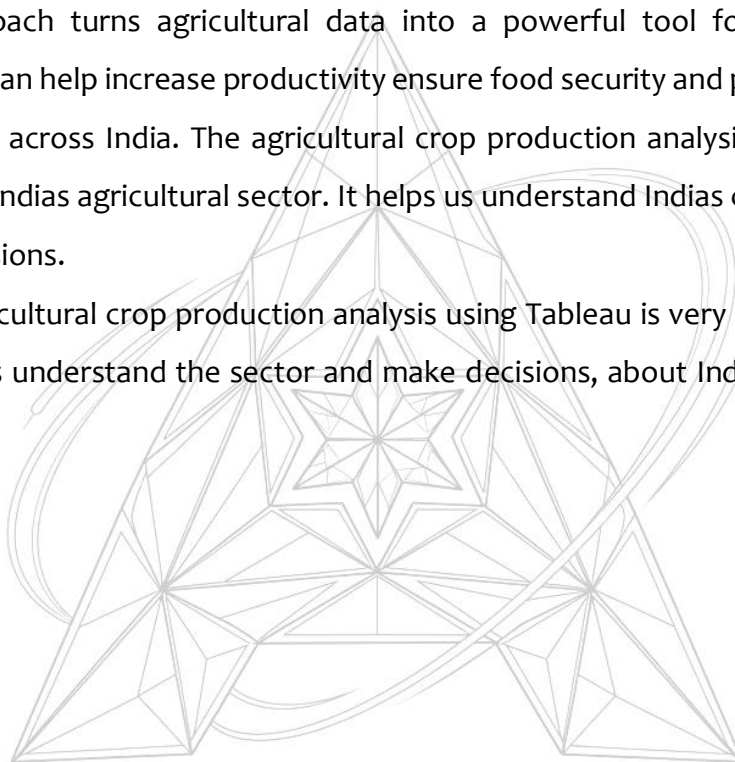
These dashboards are interactive so users can explore the data in their way. They can use filters, slicers and KPIs to find what they need. Finally, we put everything together in a storytelling dashboard, which we often share on Tableau Public so everyone can see it.

We also use maps to show where crops are grown in Indian states and districts. This helps us compare regions and find areas that are not doing well. We do analytics like calculating year-over-year growth and finding correlations between yield and external factors like rainfall and fertilizer usage.

We can even predict production based on past trends, which helps stakeholders prepare for challenges and opportunities. The dashboard is easy to use, for people who are not tech-savvy. This means farmers, policymakers and agribusiness professionals can easily get information. We update the system regularly. Make sure it can handle new data every year. We can also add crops or regions as needed.

This approach turns agricultural data into a powerful tool for making informed decisions, which can help increase productivity ensure food security and promote sustainable farming practices across India. The agricultural crop production analysis using Tableau is a step forward for Indias agricultural sector. It helps us understand Indias crop production and make better decisions.

Indias agricultural crop production analysis using Tableau is very important for Indias sector. It helps us understand the sector and make decisions, about Indias agricultural crop production.



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